Colligative Properties

Colligative properties are properties that depend on the concentration of a solute but not on its identity.

Definition: A colligative property may be defined as one which depends on the number of particles in solution and not in any way on the size or chemical nature of the particles.

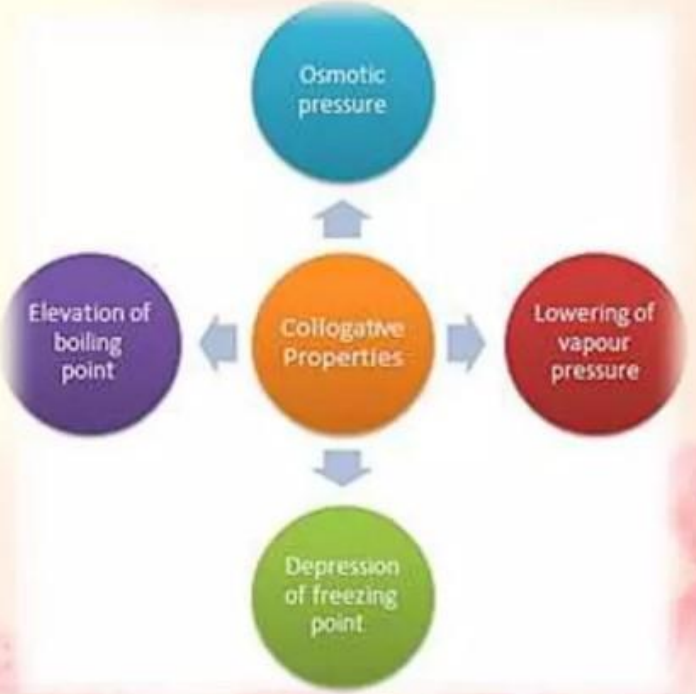
The four principal colligative properties are

(1) Lowering of the Vapour Pressure

(2) Elevation of the Boiling Point

(3) Depression of the Freezing Point

(4) Osmotic Pressure



The essential feature of these properties is that they depend *only* on the number of solute particles present in solution. Being closely related to each other through a common explanation, these have been grouped together under the class name ***Colligative Properties*** (Greek *colligatus* = Collected together).

Importance

a) Molecular mass of substances can be determined.
b) Whether a solution is iso-osmotic or not can be found.
c) The behavior of solution of electrolytes can be understood.
d) The osmotic properties of body fluids such as lacrimal fluids and blood can be evaluated.
e) Isotonic solutions can be prepared.

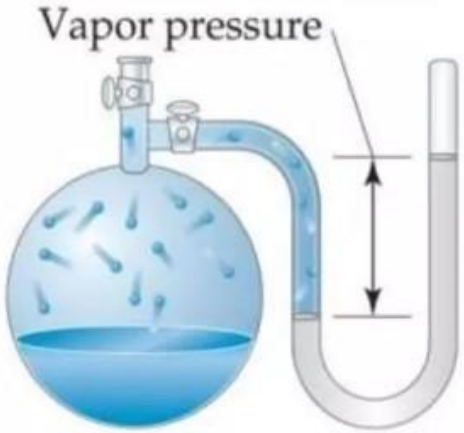
Colligative Properties – The lowering of Vapor Pressure

The lowering of vapor pressure is a decrease in the vapor pressure of a solution when a nonvolatile solute is dissolved in a solvent:

The solute particles occupy some of the surface area that would otherwise be available for solvent molecules to evaporate. This results in less solvent being able to evaporate, which lowers the vapor pressure.

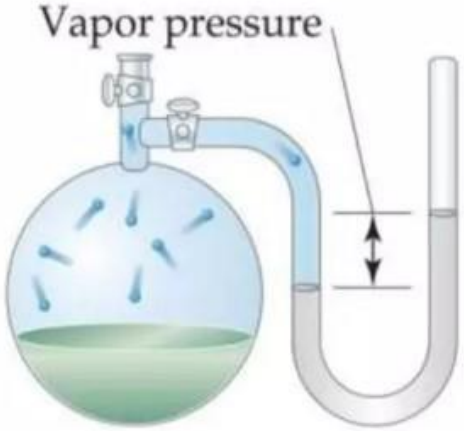
- A solvent in a closed container reaches a state of dynamic equilibrium.
- The pressure exerted by the vapor in the headspace is referred to as the vapor pressure of the solvent.
- The addition of any nonvolatile solute (one with no measurable vapor pressure) to any solvent reduces the vapor pressure of the solvent.
- Nonvolatile solutes reduce the ability of the surface solvent molecules to escape the liquid.
- Vapor pressure is reduced.
- The extent of vapor pressure lowering depends on the amount of solute.
- Raoult’s Law quantifies the amount of vapor pressure lowering that is observed.

Lowering of vapour pressure



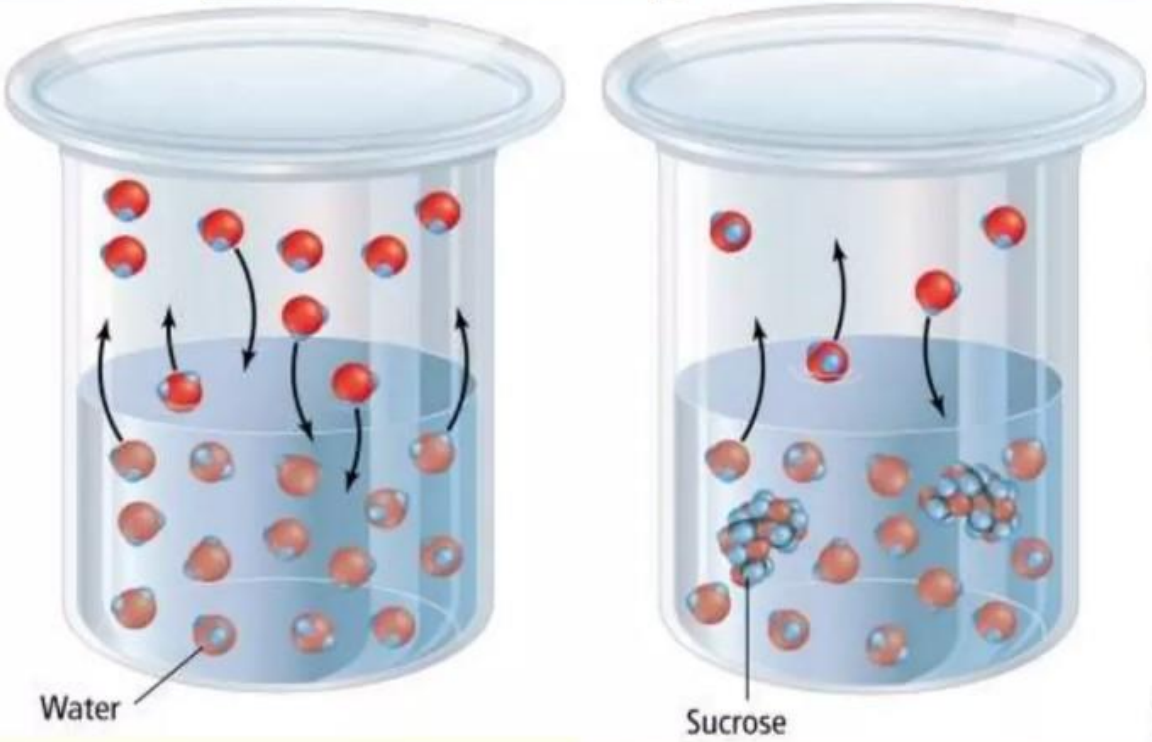
Solvent alone

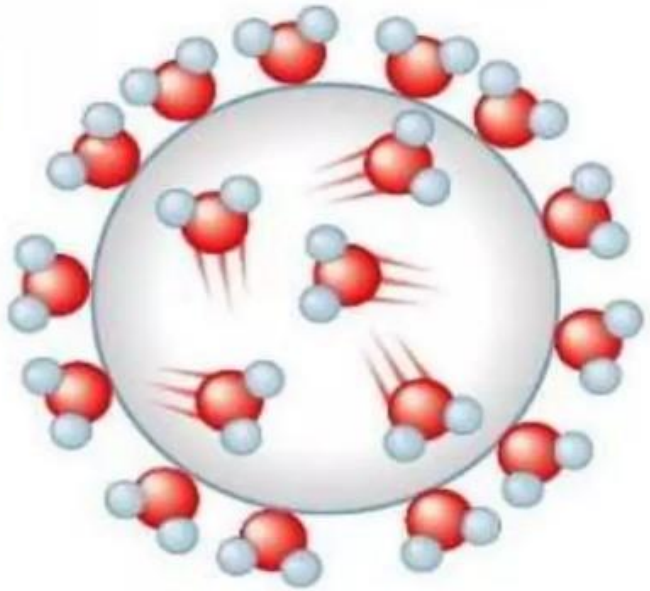
(a)



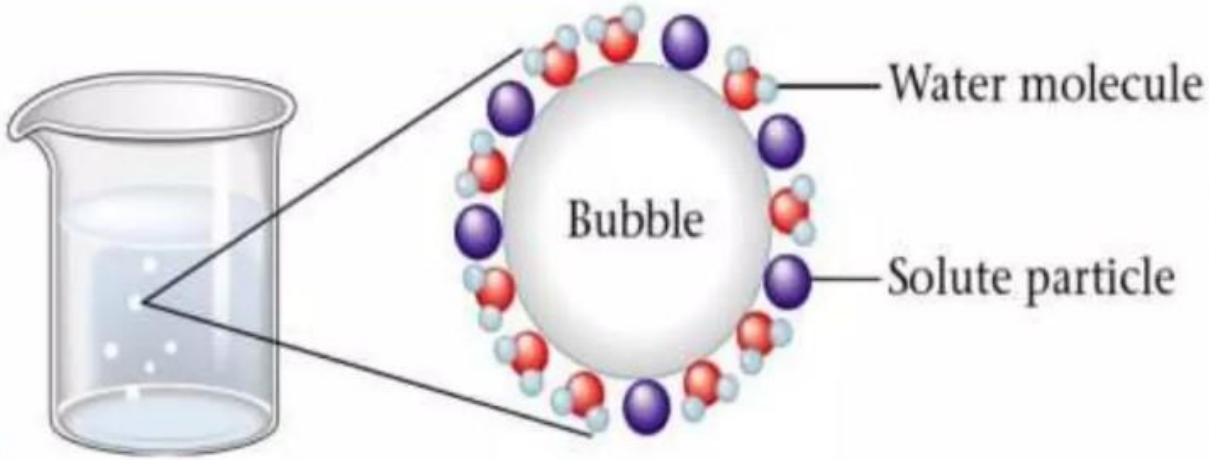
Solvent + solute

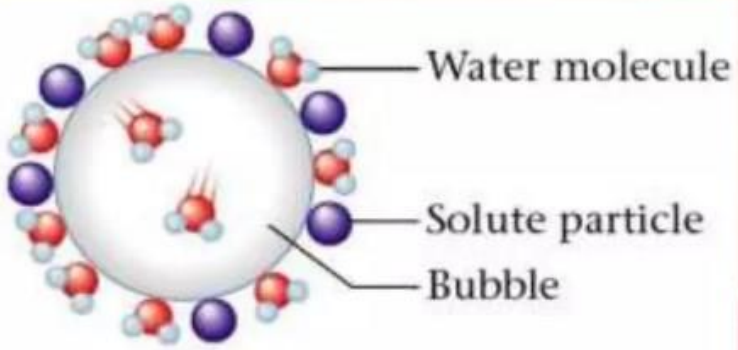
(b)





(a) Pure water





(b) Solution (contains solute)

Lowering of Vapour Pressure: Raoult's Law

The vapour pressure of a pure solvent is decreased when a non-volatile solute is dissolved in it. Raoult (1886) gave an empirical relation, connecting the relative lowering of vapour pressure and the concentration of the solute in solution. This is now referred to as the **Raoult's Law.**

It states that: **the relative lowering of the vapour pressure of a dilute solution is equal to the mole fraction of the solute present in dilute solution.**

If p is the vapour pressure of the solvent and p_s that of the solution, the lowering of vapour pressure is $(p - p_s)$. This lowering of vapour pressure relative to the vapour pressure of the pure solvent is termed the **Relative lowering of Vapour pressure.**

Thus,

Relative Lowering of Vapour Pressure $= \dfrac{p - p_s}{p}$

Therefore, Raoult's Law can be expressed mathematically in the form: $\dfrac{p - p_s}{p} = \dfrac{n}{n + N}$

where n = number of moles or molecules of solute, N = number of moles or molecules of solvent.

Derivation of Raoult's Law

Let, p is the vapour pressure of the solvent and p_s that of the solution, the vapor pressu re of the solution is directly proportional to the mole fraction of the solvent. *The vapor pressure of the* ***solution is,*** *therefore, determined by the number of molecules of the solvent present at any time in the surface which is proportional to the mole fraction.*

That is, $p_s \propto \frac{N}{n+N}$

Where N = moles of solvent and n = moles of solute.

$$p_s = k\frac{N}{n+N} \text{--------}(1)$$

Where, k =proportionality factor.

In case of pure solvent, n=0

And hence mole fraction of solvent $= \frac{N}{n+N} = \frac{N}{0+N} = 1$

Now from equation (1), the vapor pressure of the solvent $p = k$

Therefore the equation (1) assumes the form

This is Raoult's Law.

$$p_s = p\frac{N}{n+N}$$

$$\frac{p_s}{p} = \frac{N}{n+N}$$

$$1 - \frac{p_s}{p} = 1 - \frac{N}{n+N}$$

$$\frac{p - p_s}{p} = \frac{n}{n+N}$$

Determination of Molecular Mass from Vapour Pressure Lowering

The molecular mass of a non-volatile solute can be determined by measuring the lowering of vapour pressure ($p - p_s$) produced by dissolving a known weight of it in a known weight of the solvent. If in a determination *w* grams of solute is dissolved in *W* grams of the solvent, *m* and *M* are molecular masses of the solute and solvent respectively, we have:

No. of Moles of solute $(n) = \frac{w}{m}$ and No. of Moles of solvent $(N) = \frac{W}{M}$

We know that, Raoult's Law $\frac{p - p_s}{P} = \frac{n}{n + N}$

Substituting these values in the Raoult's law Equation, $\frac{p - p_s}{P} = \frac{w/m}{w/m + W/M}$ ----------(1)

Since for very dilute solution, the number of moles (molecules) of solute (w/m), is very small, it can be neglected in the denominator.

The equation (1) can now be written as $\frac{p - p_s}{P} = \frac{wM}{mW}$ ----------(2)

Knowing the experimental value of $p - p_s/p$, and the molecular mass of the solvent (M), the molecular weight of solute (m) can be calculated from (1) or (2).

- ✓ Ideal solutions are those that obey Raoult's Law.
- ✓ Real solutions show approximately ideal behavior when:
  - The solution concentration is low
  - The solute and solvent have similarly sized molecules
  - The solute and solvent have similar types of intermolecular forces.

- Raoult's Law breaks down when solvent-solvent and solute-solute intermolecular forces of attraction are much stronger or weaker than solute-solvent intermolecular forces.

Colligative Properties – Vapor Pressure

- Raoult's Law:

$$P_A = X_A P^o_A$$

where P_A = partial pressure of the solvent (A) vapor above the solution (ie with the solute)

X_A = mole fraction of the solvent (A)

P^o_A = vapor pressure of the pure solvent (A)

Example: The vapor pressure of pure water at 110°C is 1070 torr. A solution of ethylene glycol and water has a vapor pressure of 1.10 atm at the same temperature.

What is the mole fraction of ethylene glycol in the solution?